Module 18

Entity-Relationship Model/1

Milav Dabgar

www.milav.in

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Module Recap

- Predicate Calculus
- Tuple Relational and Domain Relational Calculus
- Equivalence of Relational Algebra and Relational Calculus

Module Objectives

- To understand the Design Process for Database Systems
- To study the E-R Model for real world representation

Module Outline

- Design Process
- E-R Model
  - Entity and Entity Set
  - Relationship
  - Cardinality
  - Attributes
  - Weak Entity Sets

Design Process

What is Design?

A Design:

- Satisfies a given (perhaps informal) functional specification
- Conforms to limitations of the target medium
- Meets implicit or explicit requirements on performance and resource usage
- Satisfies implicit or explicit design criteria on the form of the artifact
- Satisfies restrictions on the design process itself, such as its length or cost, or the tools available for doing the design

Abstraction

Role of Abstraction

- Disorganized Complexity results from
  - Storage (STM) limitations of human brain - an individual can simultaneously comprehend of the order of seven, plus or minus two chunks of information
  - Speed limitations of human brain - it takes the mind about five seconds to accept a new chunk of information
- Abstraction provides the major tool to handle Disorganized Complexity by chunking information
- Ignore inessential details, deal only with the generalized, idealized model of the world

Models

Model Building

- Physics: Time-Distance Equation, Quantum Mechanics
- Chemistry: Valency-Bond Structures
- Geography: Maps, Projections
- Electrical Circuits: Kirchoff's Loop Equations, Time Series Signals and FFT
- Transistor Models: Schematic Diagram, Interconnect Routing
- Building & Bridges: Drawings - Plan, Elevation, Side view, Finite Element Models
- Models are common in all engineering disciplines
- Model building follows principles of decomposition, abstraction, and hierarchy
- Each model describes a specific aspect of the system
- Build new models upon old proven models

Design Approach

Design Approach

- Requirement Analysis: Analyse the data needs of the prospective database users
  - Planning
  - System Definition
- Database Designing: Use a modeling framework to create abstraction of the real world
  - Logical Model
  - Physical Model
- Implementation
  - Data Conversion and Loading
- Testing

Design Approach (2): Database Designing

- **Logical Model**: Deciding on a good database schema
  - **Business Decision**: What attributes should we record in the database?
  - **Computer Science Decision**: What relation schema should we have and how should the attributes be distributed among the various relation schema?
- **Physical Model**: Deciding on the physical layout of the database

Design Approach (3): Database Designing: Logical Model

- **Entity Relationship Model**
  - Models an enterprise as a collection of entities and relationships
  - **Entity**: A distinguishable 'thing' or 'object' in the enterprise
    - Described by a set of attributes
  - **Relationship**: An association among multiple entities
  - Represented by an Entity-Relationship or ER Diagram
- **Database Normalization**
  - Formalize what designs are bad, and test for them

ER Model

Entity Relationship (ER) Model

ER Model: Database Modeling

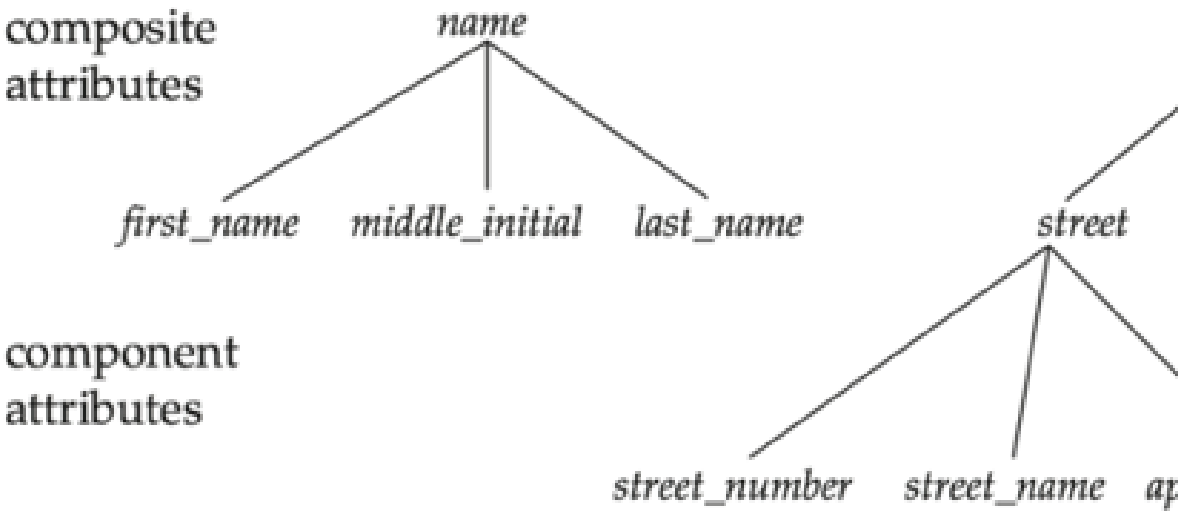
- The ER data model was developed to facilitate database design by allowing specification of an enterprise schema that represents the overall logical structure of a database
- The ER model is useful in mapping the meanings and interactions of real-world enterprises onto a conceptual schema
- The ER data model employs three basic concepts:
  - Attributes
  - Entity sets
  - Relationship sets
- The ER model also has an associated diagrammatic representation, the ER diagram, which can express the overall logical structure of a database graphically

Attributes

Attributes

- An Attribute is a property associated with and entity / entity set. Based on the values of certain attributes, an entity can be identified uniquely
- Attribute types:
  - Simple and Composite attributes
  - Single-valued and Multivalued attributes
    - Example: Multivalued attribute: phone numbers
  - Derived attributes
    - Can be computed from other attributes
    - Example: age, given date of birth
- Domain: Set of permitted values for each attribute

Attributes (2): Composite



Entity Sets

Entity Sets

- An entity is an object that exists and is distinguishable from other objects.
  - Example: specific person, company, event, plant
- An entity set is a set of entities of the same type that share the same properties.
  - Example: set of all persons, companies, trees, holidays
- An entity is represented by a set of attributes; i.e., descriptive properties possessed by all members of an entity set.
  - Example: `instructor = (ID, name, street, city, salary)`, `course = (course_id, title, credits)`
  - A subset of the attributes form a primary key of the entity set; that is, uniquely identifying each member of the set.
  - Primary key of an entity set is represented by underlining it.

Entity Sets instructor and student

76766	Crick
45565	Katz
10101	Srinivasan
98345	Kim
76543	Singh

98988	Tanaka
12345	Shankar
00128	Zhang
76543	Brown
76653	Aoi
23121	Chavez

Relationship

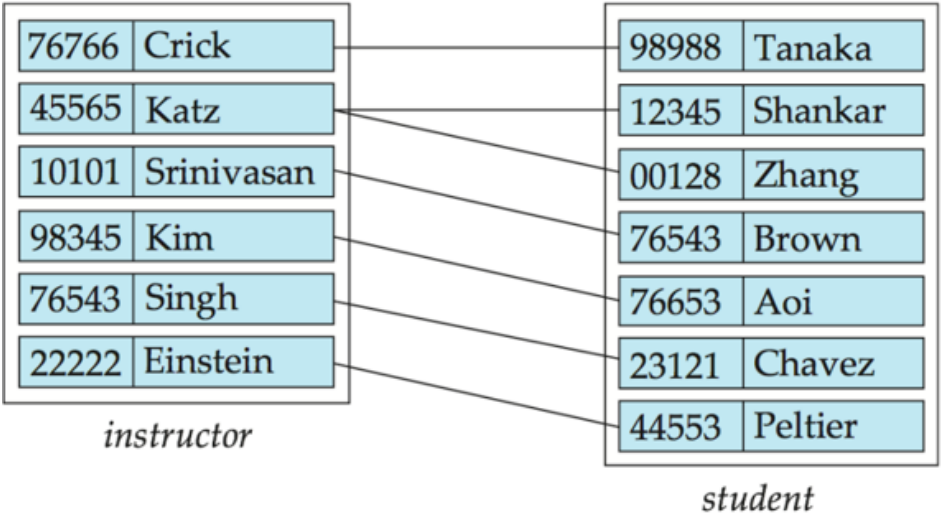
Relationship Sets

- A relationship is an association among several entities
- Example: `44553 (Peltier) -> advisor -> 22222 (Einstein)`
- A relationship set is a mathematical relation among $n \geq 2$ entities, each taken from entity sets

$$\{(e_1, e_2, \ldots e_n) \mid e_1 \in E_1, e_2 \in E_2, \ldots, e_n \in E_n\}$$

  where $(e_1, e_2, \ldots e_n)$ is a relationship.
- Example: $(44553, 22222) \in$ advisor

Relationship Set (2) advisor



Relationship Sets (3)

- An attribute can also be associated with a relationship set.
- For instance, the `advisor` relationship set between entity sets `instructor` and `student` may have the attribute `date` which tracks when the student started being associated with the advisor

Relationship Set (4): Degree

- Binary relationship
  - involves two entity sets (or degree two).
  - most relationship sets in a database system are binary.
- Relationships between more than two entity sets are rare. Most relationships are binary
  - Example: students work on research projects under the guidance of an instructor.
  - relationship `proj_guide` is a ternary relationship between `instructor`, `student`, and `project`

Attributes (3): Redundant

- Suppose we have entity sets:
  - `instructor`, with attributes: ID, name, dept_name, salary
  - `department`, with attributes: dept_name, building, budget
- We model the fact that each instructor has an associated department using a relationship set `inst_dept`
- The attribute `dept_name` appears in both entity sets. Since it is the primary key for the entity set `department`, it replicates information present in the relationship and is therefore redundant in the entity set `instructor` and needs to be removed
- BUT: When converting back to tables, in some cases the attribute gets reintroduced, as we will see later

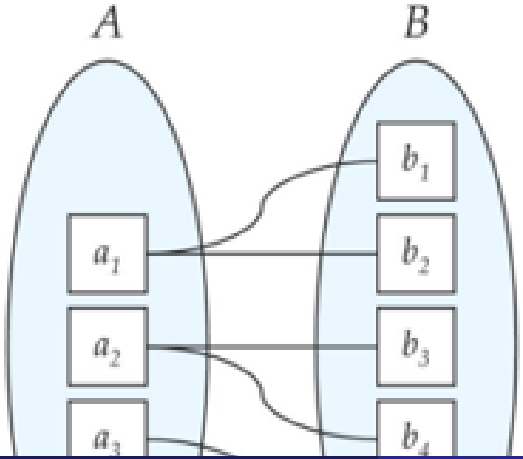
Cardinality

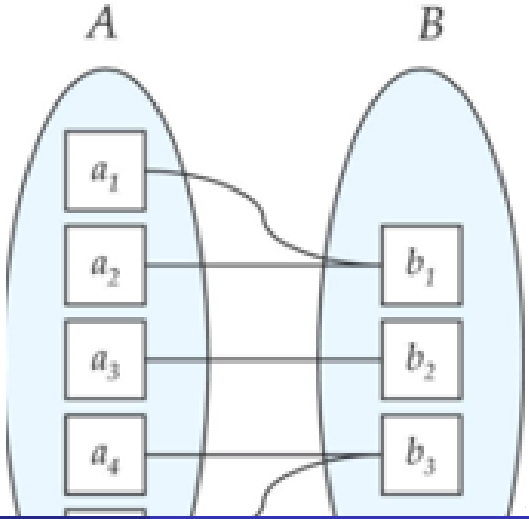
Mapping Cardinality Constraints

- Express the number of entities to which another entity can be associated via a relationship set.
- Most useful in describing binary relationship sets.
- For a binary relationship set the mapping cardinality must be one of the following types:
  - One to one
  - One to many
  - Many to one
  - Many to many

Mapping Cardinalities

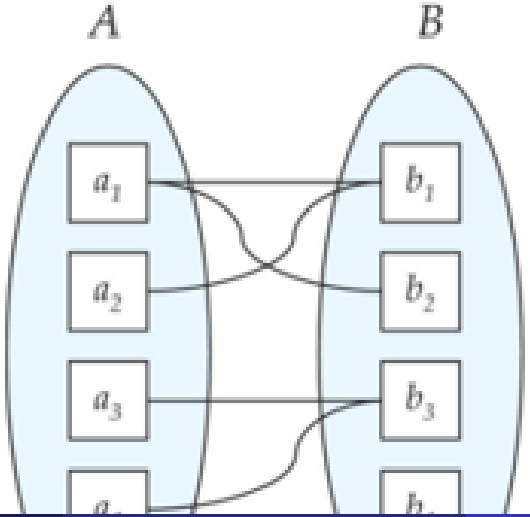
Note: Some elements in A and B may not be mapped to any elements in the other set





Mapping Cardinalities (2)

Note: Some elements in A and B may not be mapped to any elements in the other set



Weak Entity Sets

Weak Entity Sets

An entity set may be of two types:

- Strong entity set
  - A strong entity set is an entity set that contains sufficient attributes to uniquely identify all its entities
  - In other words, a primary key exists for a strong entity set
  - Primary key of a strong entity set is represented by underlining it
- Weak entity set
  - A weak entity set is an entity set that does not contain sufficient attributes to uniquely identify its entities
  - In other words, a primary key does not exist for a weak entity set
  - However, it contains a partial key called as a discriminator
  - Discriminator can identify a group of entities from the entity set
  - Discriminator is represented by underlining with a dashed line

Weak Entity Sets (2)

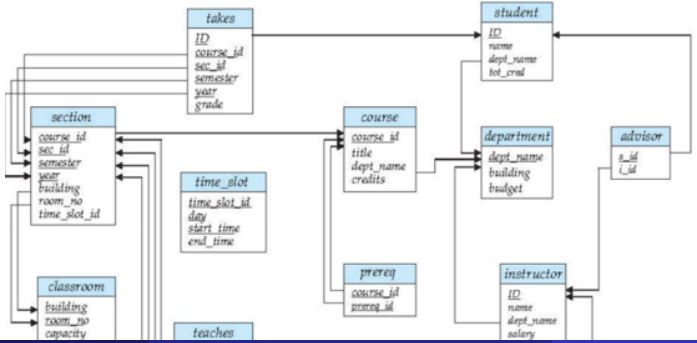
- Since a weak entity set does not have primary key, it cannot independently exist in the ER Model
- It features in the model in relationship with a strong entity set. This is called the identifying relationship
- Primary Key of Weak Entity Set
  - The combination of discriminator and primary key of the strong entity set makes it possible to uniquely identify all entities of the weak entity set
  - Thus, this combination serves as a primary key for the weak entity set.
  - Clearly, this primary key is not formed by the weak entity set completely.
  - Primary Key of Weak Entity Set = Its own discriminator + Primary Key of Strong Entity Set
- Weak entity set must have total participation in the identifying relationship. That is all its entities must feature in the relationship

Weak Entity Sets (3): Example

- Strong Entity Set: Building (building no, building name, address). building no is its primary key
- Weak Entity Set: Apartment (door no, floor). door no is its discriminator as door no alone can not identify an apartment uniquely. There may be several other buildings having the same door number
- Relationship: BA between Building and Apartment
- By total participation in BA, each apartment must be present in at least one building
- In contrast, Building has partial participation in BA only as there might exist some buildings which has no apartment
- Primary Key: To uniquely identify any apartment
  - First, building no is required to identify the particular building
  - Second, door no of the apartment is required to uniquely identify the apartment
  - Primary key of Apartment = Primary key of Building + Its own discriminator = building no + door no

Weak Entity Sets (4): Example

- Consider a `section` entity, which is uniquely identified by a `course_id`, `semester`, `year`, and `sec_id`.
- Clearly, `section` entities are related to `course` entities. Suppose we create a relationship set `sec_course` between entity sets `section` and `course`.
- Note that the information in `sec_course` is redundant, since `section` already has an attribute `course_id`, which identifies the course with which the section is related.



Module Summary

- Introduced the Design Process for Database Systems
- Elucidated the E-R Model for real world representation with entities, entity sets, attributes, and relationships

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